CANARA COLLEGE LIBRARY MANGALURU-3

Reg. No. | | | | | | | | | | | |

BCACACS 402

Fourth Semester B.C.A. Degree Examination, June/July 2026
(SEP) (2025-26 Batch Onwards)
ADVANCED JAVA AND J2EE

Time : 3 Hours Max. Marks : 80

***Note** : Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit of Part – **B**.*

PART – A

1. a) What is an enumeration ? How enumeration can be created ? **(10×2=20)**
   b) List any two retention policies with its purpose.
   c) What is persistence in JavaBeans ?
   d) Write the differences between hasNext() and next() method.
   e) Write any two uses of Generics.
   f) What are two responsibilities of the view component in MVC ?
   g) What is the difference between equals() and == ?
   h) How does StringBuilder differ from StringBuffer ?
   i) What are the roles of the Stub and Skeleton objects in RMI ?
   j) What is ServletConfig Interface ? Mention any 2 methods.
   k) With syntax write the purpose of getParameter() method.
   l) What is Cookie ? How it is helpful ?

PART – B

Unit – I **(4×15=60)**

2. a) Java enumerations are class types. Explain with an example.
   b) How do you specify a default value for an annotation member ? Explain with suitable example.
   c) What are the steps to be followed in creating a new Bean ? **(5+5+5)**

3. a) Demonstrate the usage of the value Of() and values() methods with an example.
   b) What is autoboxing and unboxing ? Explain with an example.
   c) Write short note on the following :
      i) Bound and constrained properties
      ii) Persistence. **(5+5+5)**

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Unit – II

4. a) List any five methods of collection interface with its purpose.
   b) With an example explain the usage of ArrayList.
   c) Write a Java program that demonstrates the usage of a custom Comparator for sorting strings in reverse order. **(5+4+6)**

5. a) List any five methods of List interface with its purpose.
   b) What is an iterator ? Explain with an example.
   c) Explain the roles and responsibilities of the Model, View and Controller components in the MVC architecture. **(5+4+6)**

Unit – III

6. a) With syntax and example explain the following StringBuffer class methods
      i) . insert()
      ii) . deleteCharAt()
   b) With an example explain the four methods of StringBuffer class.
   c) Explain the concept of Remote Procedure Calls (RPC) and its working mechanism with its five key elements. **(5+4+6)**

7. a) With syntax and example explain the following String class methods
      i) . split()
      ii) . regionMatches()
   b) Outline the key steps involved in developing a basic Remote Method Invocation (RMI) application.
   c) Explain the Advantages and Disadvantages distributed computing. **(5+5+5)**

Unit – IV

8. a) Explain the three key methods in the lifecycle of a servlet (init(), service(), destroy()).
   b) Write the usage of any four methods of servletRequest interface.
   c) Write the steps to associate database with JDBC/ODBC bridge. **(6+4+5)**

9. a) Explain the key stages involved in the lifecycle of a Java Server Page (JSP).
   b) Write the usage of any four methods of servletResponse interface.
   c) Explain the steps to connect to the database in java. **(6+4+5)**

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